One photo becomes **a buildable nail set**

idala is a production web service where nail salon owners upload inspiration photos — an outfit, a tattoo flash sheet, a necklace — and receive photorealistic press-on nail set designs that a human artist can actually craft by hand. Solo-built: product, multi-model AI pipeline, payments, and deployment.

ROLE
Solo developer & designer

STACK
Next.js 15 · TypeScript · Gemini · GPT-Image · Neon Postgres · Redis · Paddle

STATUS
Live on Vercel, paid product (₩9,900 / 8 runs)

THE PIPELINE — FOUR STAGES, THREE MODELS

Vision analysis 01
Gemini reads the photos into a structured design brief (JSON schema): palette, surface finish, motif vocabulary, and one **hero motif** — the single strongest pictorial element.

structured output · vision

Composition planning 02
An LLM expands the brief into three 10-tip plans (balanced / bold / quiet) with an enforced hierarchy: hero on 1–2 tips, supporting tips, 2–3 rest tips.

LLM planning · deterministic fallback

Image generation 03
GPT-Image renders each plan behind a provider abstraction (Gemini / Seedream / GPT-Image swappable; mock mode makes the test suite cost $0).

provider abstraction

LLM-as-judge 04
A vision judge scores every render on 8 criteria — tip count, hero hierarchy, part counts, palette fidelity, physical buildability. Failures trigger one retry; the better render wins.

auto-retry · 8-point rubric

SELECTED OUTPUT — GENERATED END-TO-END FROM USER PHOTOS



Winged Heart French
Hero-motif hierarchy: the heart owns exactly one tip



Glass Tears
Sculpted-gel translation of a jewelry reference



Olive Stiletto
AI try-on shot — preview before committing

409
unit tests, incl. prompt snapshot guards

8-criteria
vision rubric computed in code, not vibes

3 layers
of quota defense (account / IP / global)

62 sets
published in the public archive (/looks)

PROMPT ENGINEERING AS AN EMPIRICAL DISCIPLINE

- **Vocabulary control.** Every noun gets drawn literally — "charm" renders a hanging pendant, "anchor" renders ⚓. The system bans trap words and validates plans against them, discarding violating sets before they cost an image.
- **Positive phrasing & explicit counts.** Negative instructions leak; constraints are rewritten as what IS ("Exactly one tip carries… every other tip is painted gel only"). A regression-tested prompt baseline guards against drift.
- **Materiality → technique translation.** The analyzer doesn't describe motifs, it translates their physics: a necklace becomes a draped pearl arrangement, woven fabric becomes tone-on-tone raised gel, watercolor becomes an airbrushed wash.
- **Hero-motif hierarchy.** Uniform-density sets read as "AI-generated" to professionals. The pipeline extracts one hero motif and enforces a cast: 1–2 hero tips, supporting tips, quiet rest tips — verified by the judge.
- **Finish fidelity.** A matte illustration reference yields a matte velvet set; a glossy jewelry reference yields glass gloss. The variant axis is composition, never a style costume that overrides the photo.
- **Calibrated judging.** The vision model only reports observations (counts, booleans); verdicts are computed in code. Expected part counts are parsed from the brief itself, so painted-only references are never falsely penalized.

ITERATION, MEASURED — SAME INSPIRATION PHOTOS, ONE WEEK APART



Before — style-forced
A fixed "glass jelly" style overrode the reference's matte illustration texture.



After — photo-faithful
The same photos now yield a matte airbrushed set carrying the reference's stamps and hand-drawn motifs.

PRODUCTION ENGINEERING

- **Money-safe quotas.** Each generation is real cost, so credits use a reserve → commit / refund protocol on Redis: failed runs never consume a customer's paid credits, and a reaper refunds reservations orphaned by platform timeouts.
- **Abuse defense in depth.** Per-account, per-IP and global daily limits — account-only limits fall to free Google accounts; the global image cap is the hard ceiling on daily spend.
- **Payments without an entity.** Paddle as merchant of record (KakaoPay & Korean cards), webhook + direct transaction verification so a late webhook can't lose a purchase.
- **Security posture.** Signed blob URLs derived server-side and path-bound (verified: cross-path access returns 403); session tokens gate the generation API so the paid analyze step can't be bypassed.
- **Testing culture.** 409 unit tests; prompts are snapshot-guarded; mock providers make the full pipeline testable at zero API cost; judge rubric changes ship with fixture-level tests.
- **Validation-driven product.** Market re-validated three times with structured evidence docs; the current target (Korean salon owners) and pricing came from pain-signal mining, not intuition.